POWER-1

DCIN:12-20V

12V 27A

开关S1，整板下电

底板12V电源
12V 27A

	STANDBY	PWR_ON
J0102	0	1

VCC_BAT_AON-->STANDBY_3.3V

核心板5V电源

5V 15A

Rsh

VFB=1.0V

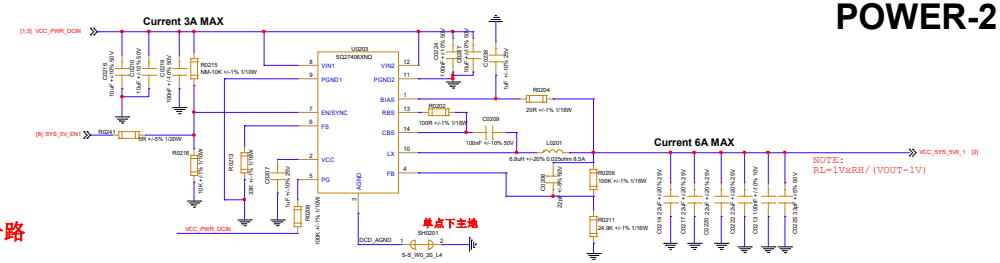
Imax = 80mV/Rsh
Freq= 414KHz --> Rposc=69.8Kohm

For Connector

Quectel Wireless Solutions	
PROJECT: QSC8910XP-AP-EVB	VER: V1.2
DRAWN BY: Cyrus	CHECKED BY: Terence
DATE: Monday, May 04, 2009	SIZE: A1
	SHEET: 1 OF 21

POWER-2

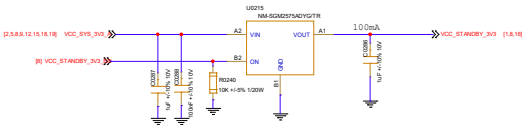
底板5V电源第一路



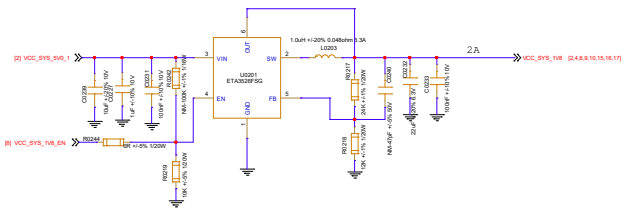
VCC_PWR_DCIN-->5V_SYS_BUCK

Vout = 1 x (R0209/ R0211 + 1) = 5.02V

预留核心板休眠电源

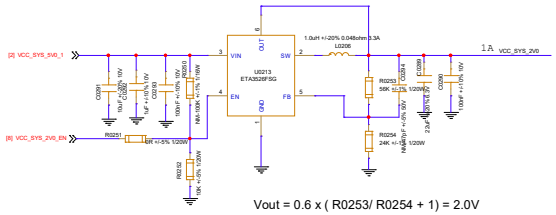


底板1.8V电源



Vout = 0.6 x (R0217/ R0218 + 1) = 1.8 V

VCC_SYS_5V0_1-->VCC_SYS_1V8



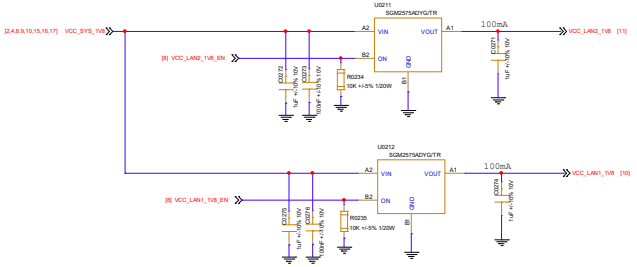
Vout = 0.6 x (R0253/ R0254 + 1) = 2.0V

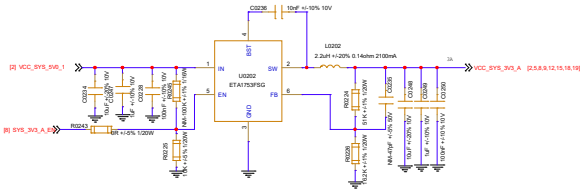
网口PHY 2V电源

VCC_SYS_5V0-->VCC_SYS_2V0

VFB=0.606V
FREQ=450KHz

网口PHY 1.8V电源负载开关

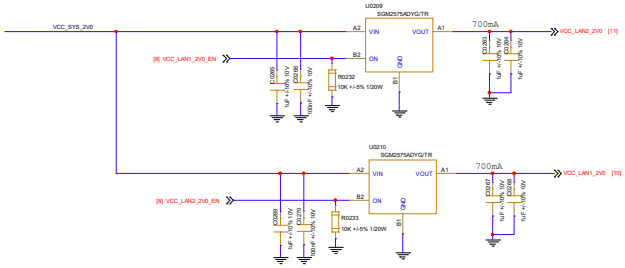




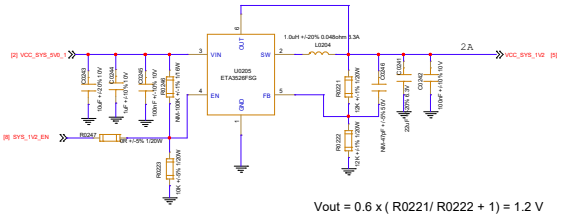
Vout = 0.806 x (R0224/ R0226 + 1) = 3.304 V

VCC_SYS_5V0_1-->VCC_SYS_3V3_A

底板3.3V电源第一路



网口PHY 2V电源负载开关

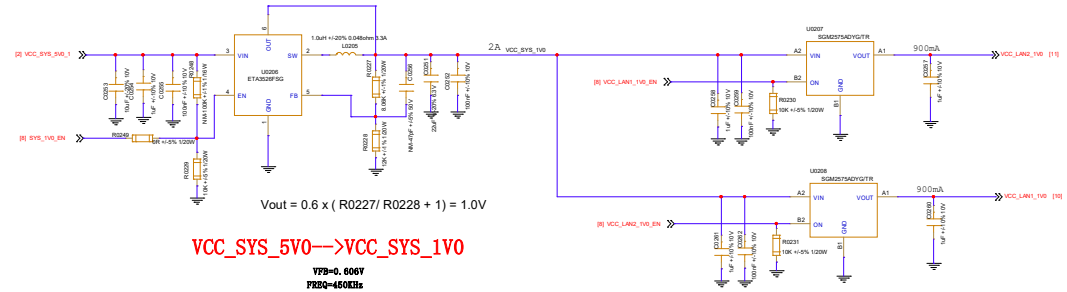


Vout = 0.6 x (R0221/ R0222 + 1) = 1.2 V

VCC_SYS_5V0-->VCC_SYS_1V2

网口PHY 1.2V电源

网口PHY 1V电源



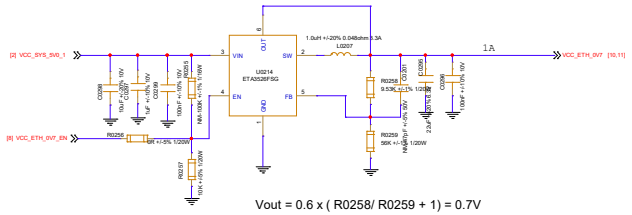
Vout = 0.6 x (R0227/ R0228 + 1) = 1.0V

VCC_SYS_5V0-->VCC_SYS_1V0

VFB=0.606V
FREQ=450KHz

网口PHY 1V电源负载开关

网口PHY 0.7V电源



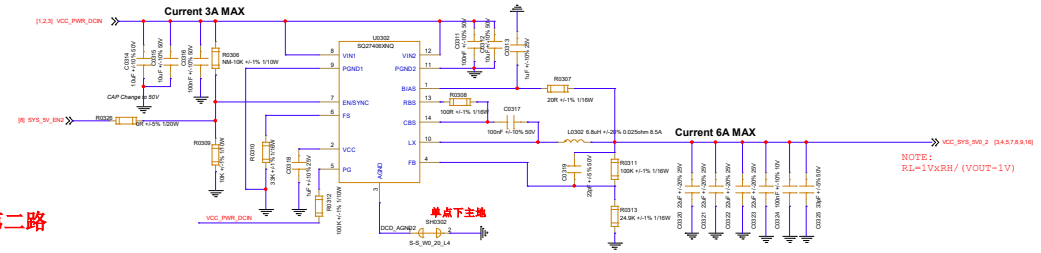
Vout = 0.6 x (R0258/ R0259 + 1) = 0.7V

VCC_SYS_5V0-->VCC_ETH_0V7

VFB=0.606V
FREQ=450KHz

Quectel Wireless Solutions			
PROJECT	QSC801XP-AP-EVB	VER	V1.2
DRAWN BY	Cyrus	CHECKED BY	Terence
SIZE A1	DATE Monday, May 04, 2026	SHEET 2	OF 21

POWER-3

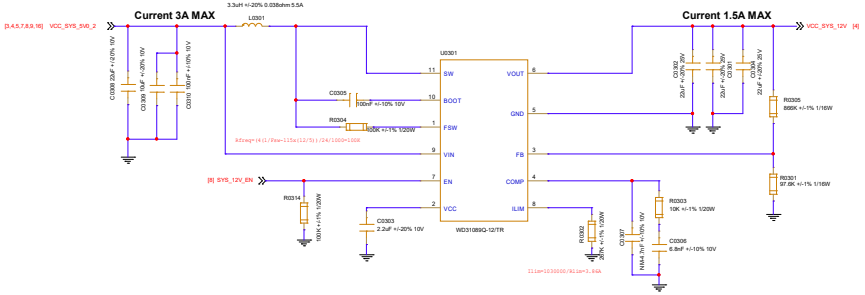


底板5V电源第二路

Vout = 1 x (R0311/ R0313 + 1) = 5.02V

VCC_PWR_DCIN —> VCC_SYS_5V0_2

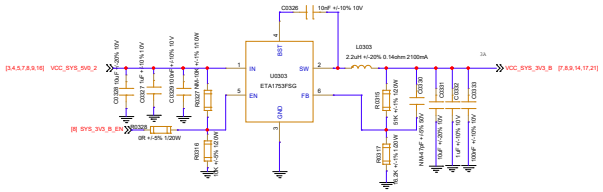
风扇12V电源



Vout = 1.212 x (R0305/ R0301 + 1) = 11.97V

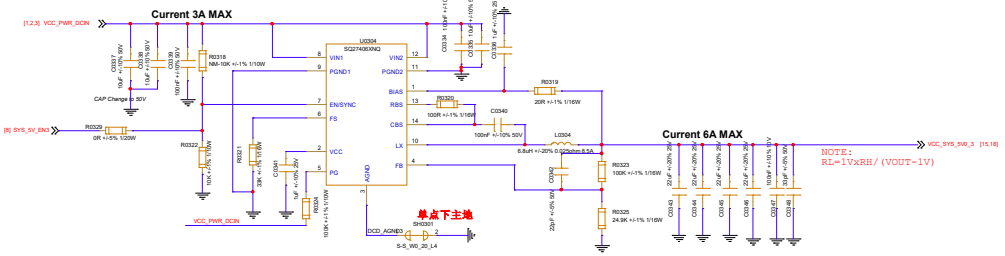
VCC_SYS_5V0_2—>VCC_SYS_12V BOOST

底板3.3V电源第二路



Vout = 0.806 x (R0315/ R0317 + 1) = 3.304 V

底板5V电源第三路

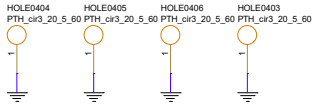


Vout = 1 x (R0323/ R0325 + 1) = 5.02V

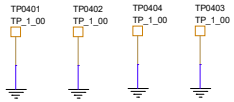
VCC_PWR_DCIN —> VCC_SYS_5V0_3

Quectel Wireless Solutions			
PROJECT	QSC801XP-AP-EVB	VER	V1.2
DRAWN BY	Cyrus	CHECKED BY	Terence
SIZE A1	DATE Monday, May 04, 2026	SHEET 3	OF 21

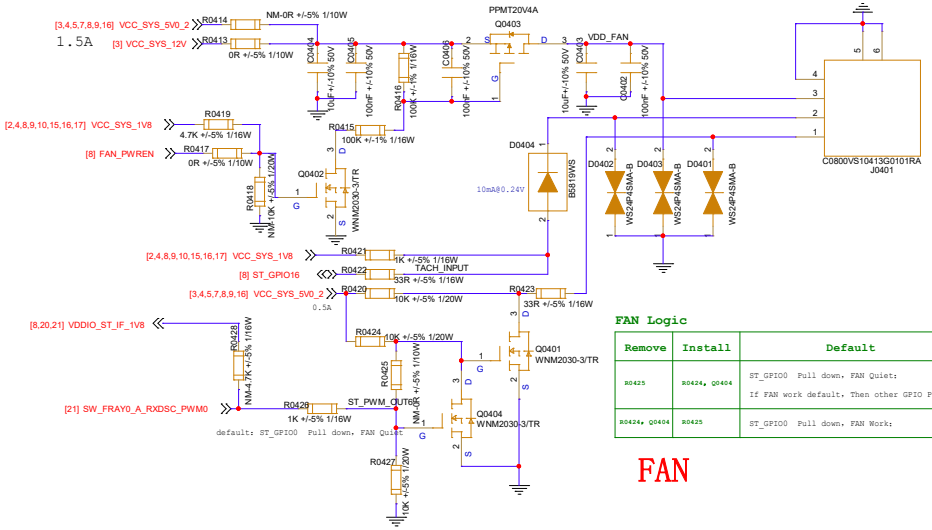
安装孔



到位检测地



风扇

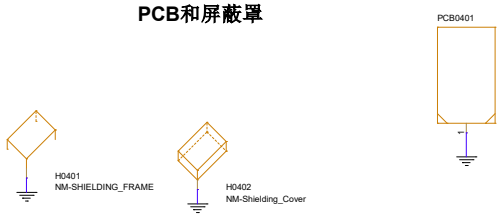


FAN Logic

Remove	Install	Default
R0425	R0424, Q0404	ST_GPIO0 Pull down. FAN Quiet; If FAN work default. Then other GPIO Pull up default
R0424, Q0404	R0425	ST_GPIO0 Pull down. FAN Work;

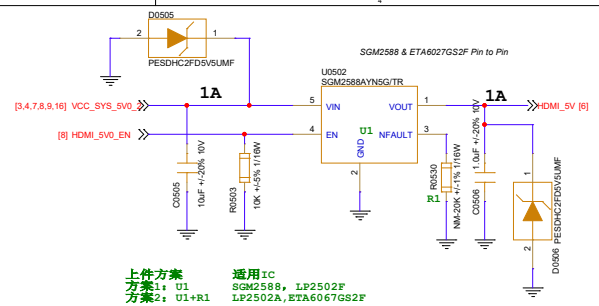
FAN

PCB和屏蔽罩



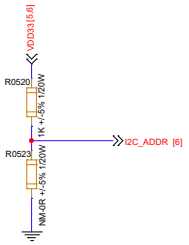
Quectel Wireless Solutions			
PROJECT: QSC801XP-AP-EVB			VER: V1.2
DRAWN BY: Cyrus			CHECKED BY: Terence
SIZE: A2	DATE: Monday, May 04, 2026	SHEET: 4	OF: 21

HDMI 5V POWER



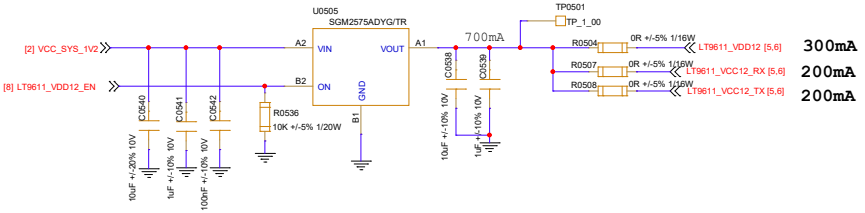
上件方案　　适用IC
方案1: U1　　SGM2588, LP2502F
方案2: U1+R1　　LP2502A,ETA6067GS2F

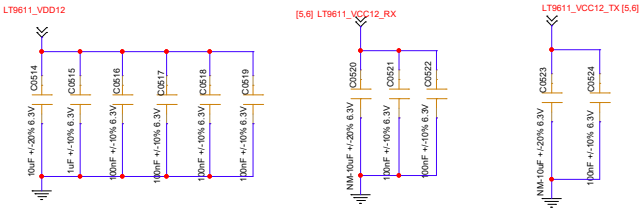
请注意，R1的阻值需要根据IC需求计算得出，请勿混用



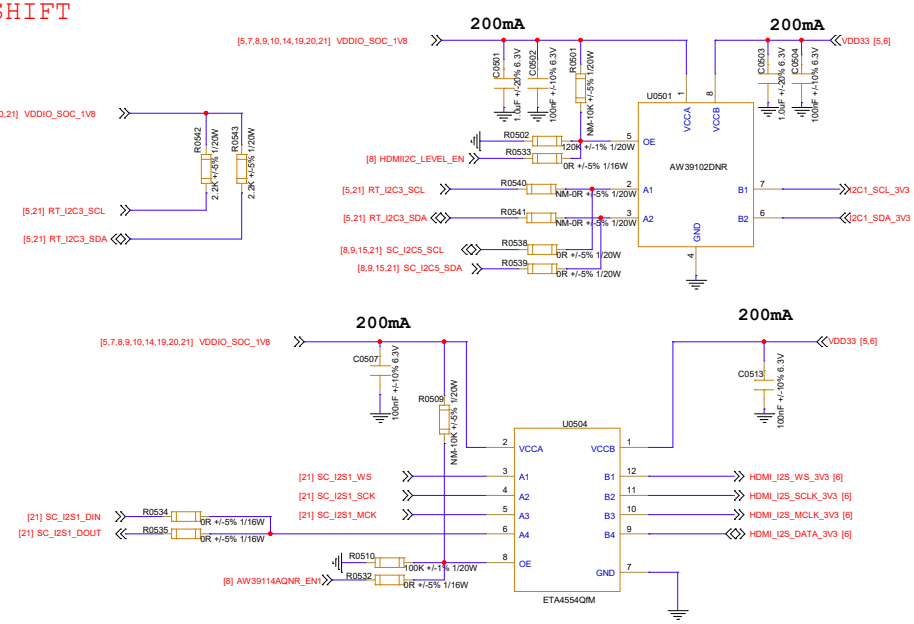
Note:
1.2V power should be set up at least 1ms later than 3.3V power, the internal reset signal should be released after 1.2V power is ready.
External RESET# signal should be set up at least 1ms later than 1.2V.
The 1.2V power supply enable pin can be connected to the 3.3V power supply.

LT9611 1V2

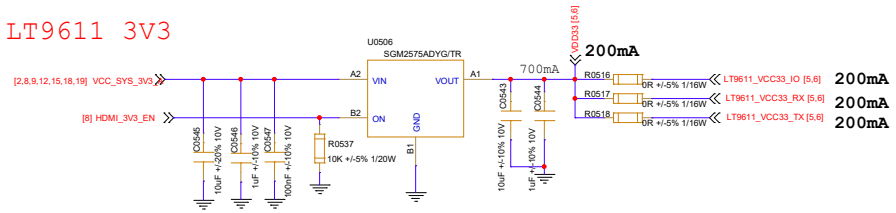


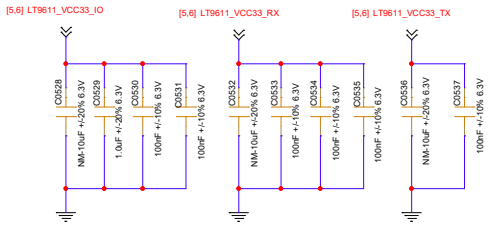


LEVEL SHIFT

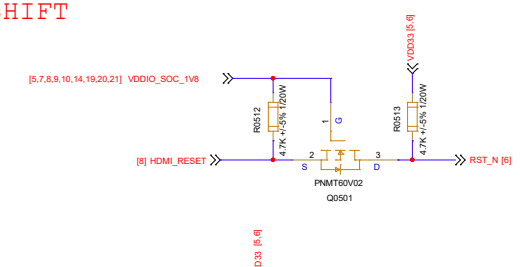


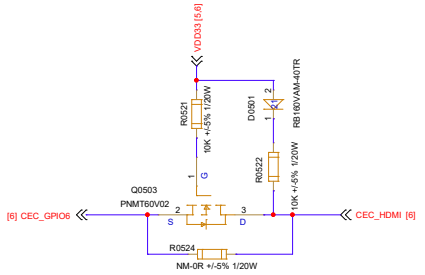
LT9611 3V3





LEVEL SHIFT

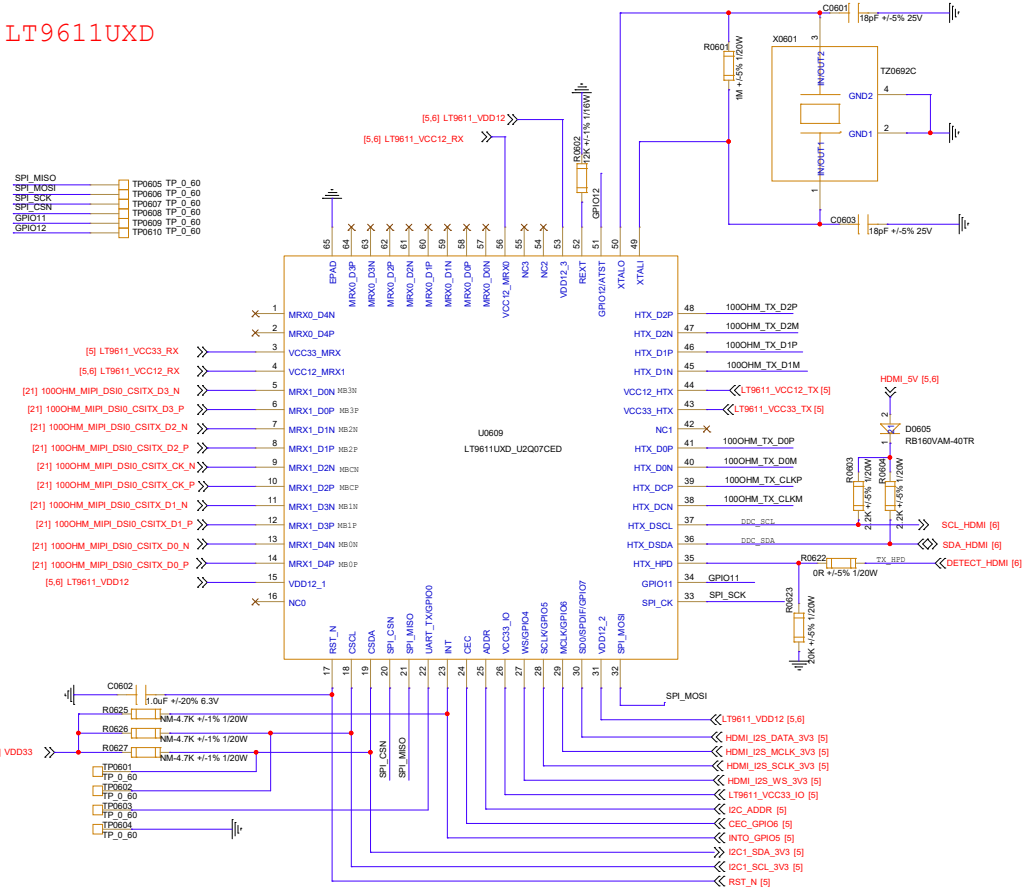




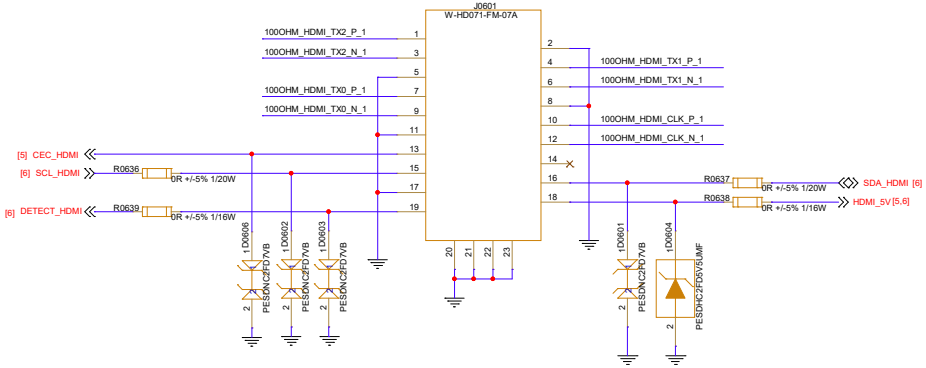
CEC　为5V d电压防止5V电压倒灌到3.3V

Quectel Wireless Solutions		
PROJECT: QSC801XP-AP-EVB		VER: V1.2
DRAWN BY: Cyrus	CHECKED BY: Terence	SIZE: A2
DATE: Monday, May 04, 2026	SHEET: 5 OF 21	

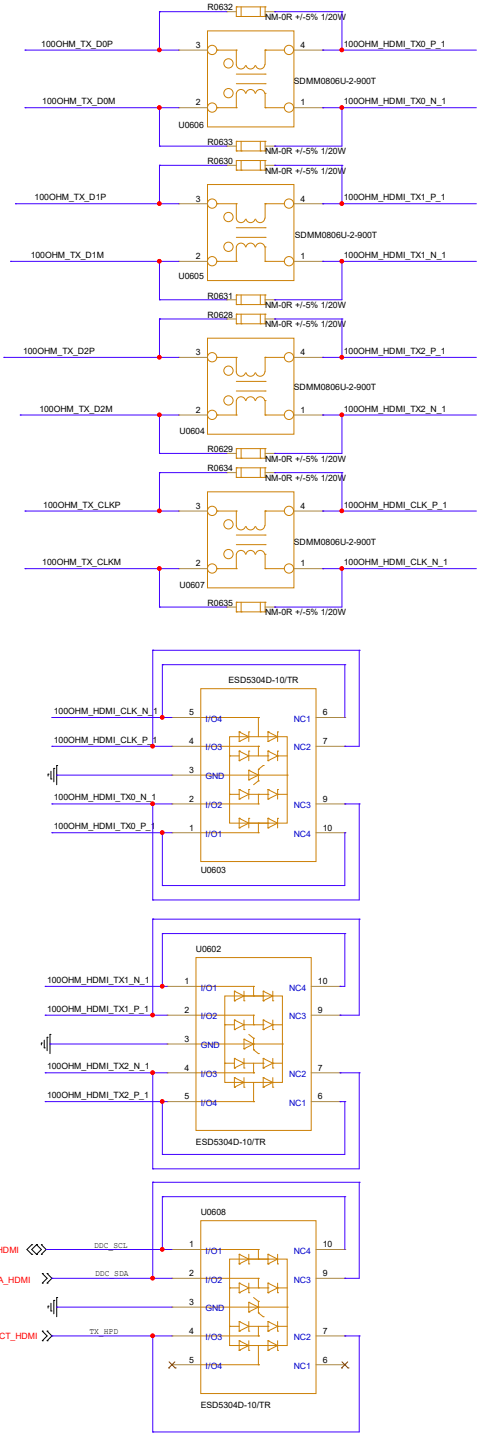
LT9611UXD



HDMI CONNECTER

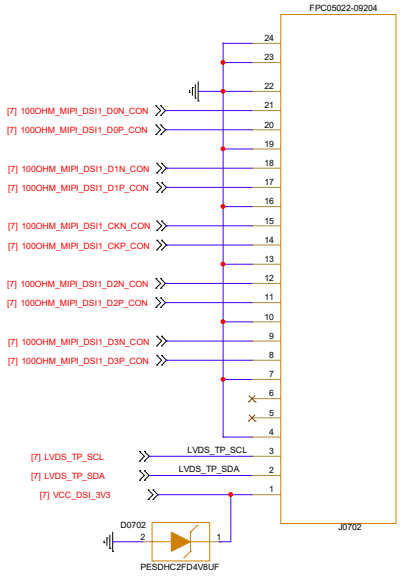


EMC Protection

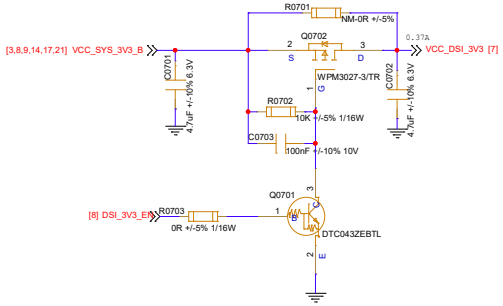


Quectel Wireless Solutions			
PROJECT	QSC801XP-AP-EVB	VER	V1.2
DRAWN BY	Cyrus	CHECKED BY	Terence
SIZE	A2		
DATE	Monday, May 04, 2026	SHEET	6 OF 21

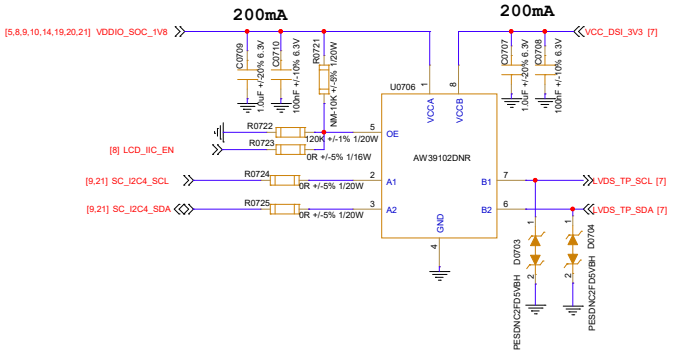
MIPI LCD CONNECTER



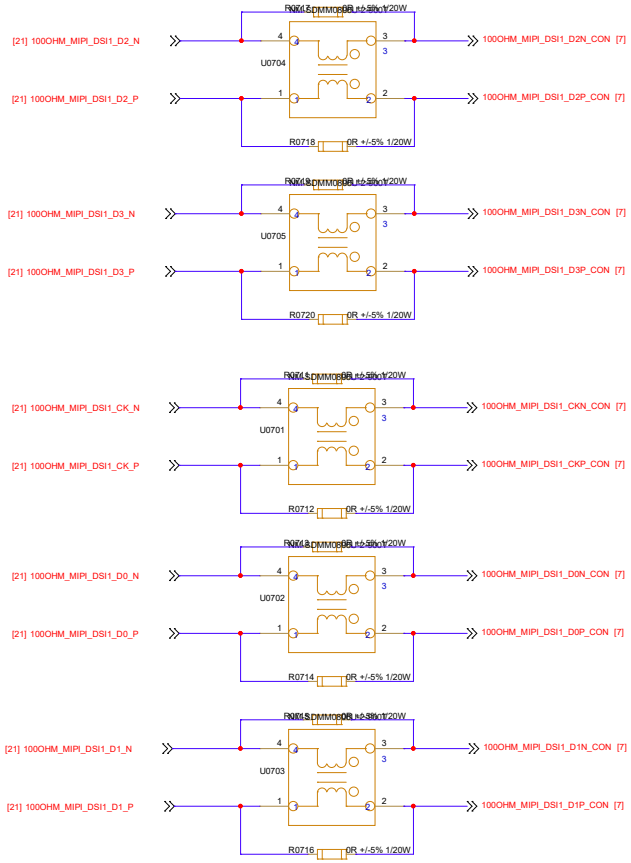
LCD 3.3供电



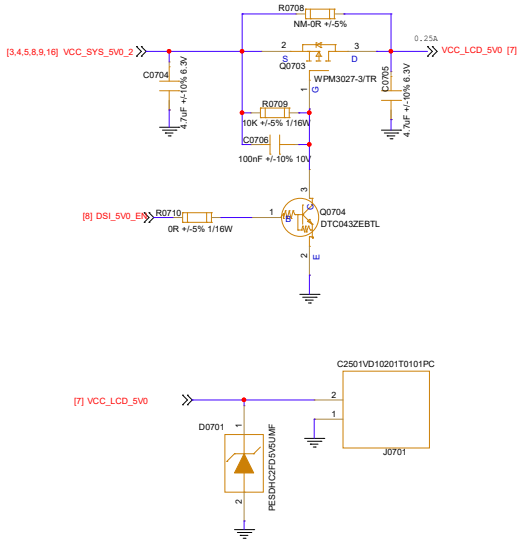
LEVEL SHIFT



EMC Protection

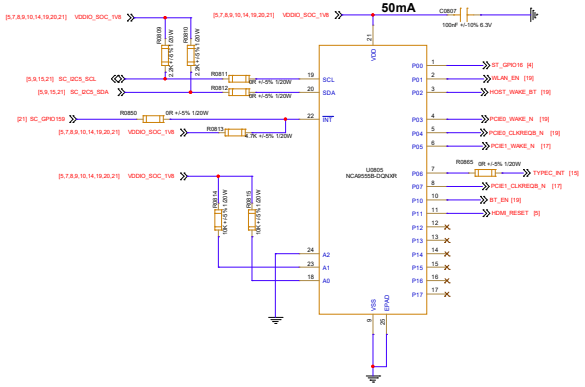


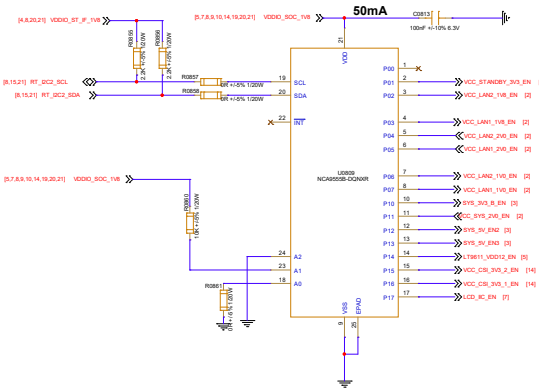
LCD 5V供电

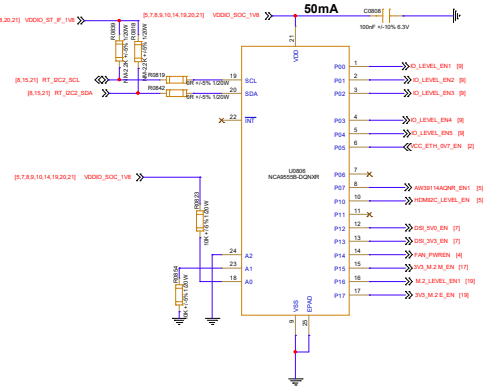


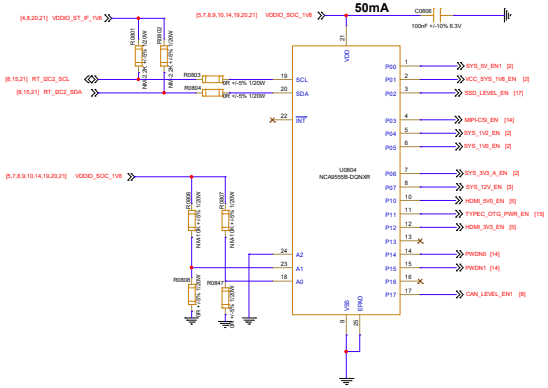
Quectel Wireless Solutions		
PROJECT: QSC801XP-AP-EVB		VER: V1.2
DRAWN BY: Cyrus	CHECKED BY: Terence	SIZE: A2
DATE: Monday, May 04, 2026	SHEET: 7 OF 21	

IO Expansion

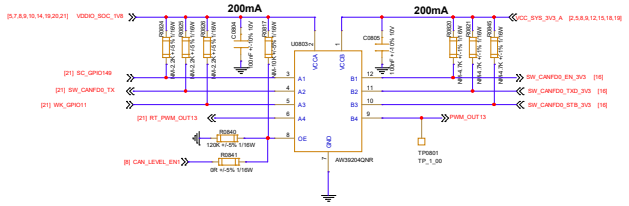




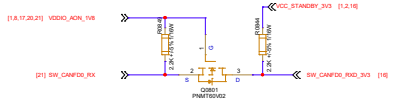




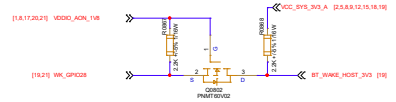
LEVEL SHIFT



AON

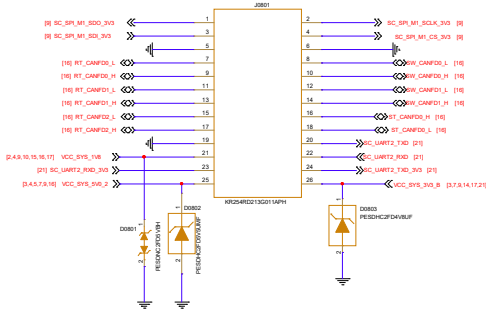


CAN WAKE UP SOC



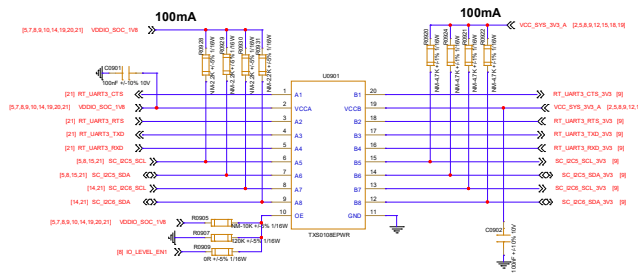
BT WAKE UP SOC

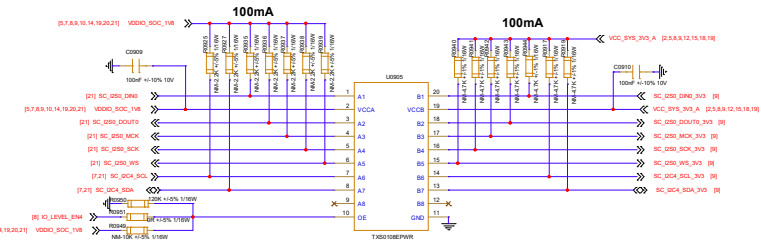
26PIN Header

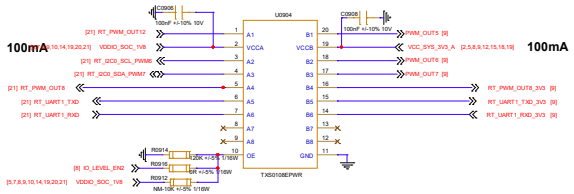


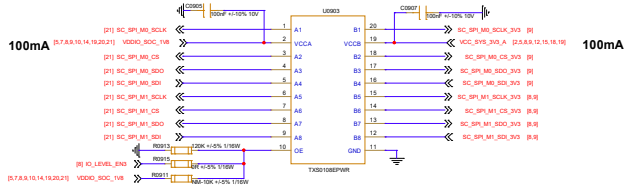
Quectel Wireless Solutions			
PROJECT: QSC801XP-AP-EVB		VER: V1.2	
DRAWN BY: Cyrus	CHECKED BY: Terence		SIZE: A0
DATE: Monday, May 04, 2026		SHEET: 8 OF 21	

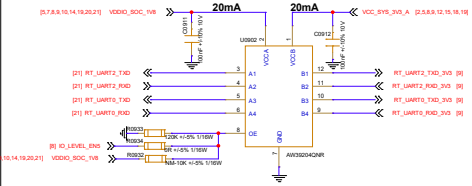
LEVEL SHIFT



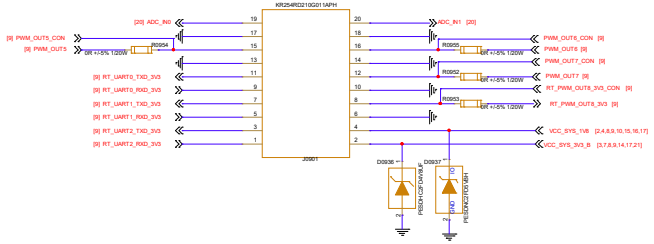




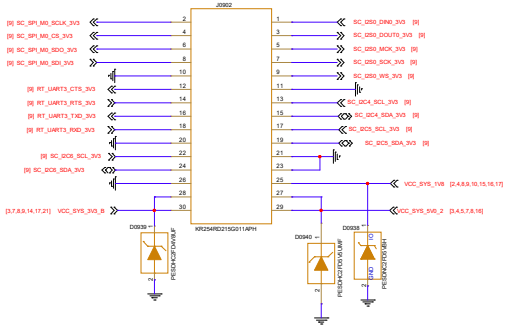




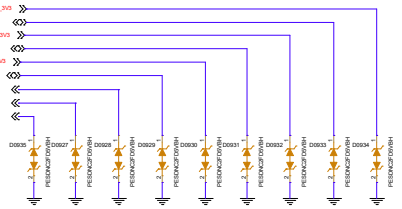
20PIN Header

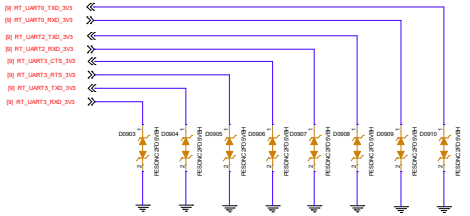


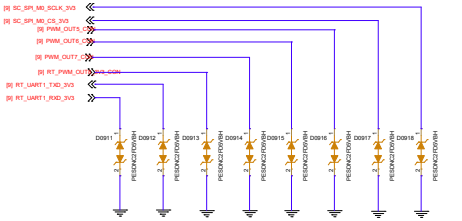
30PIN Header

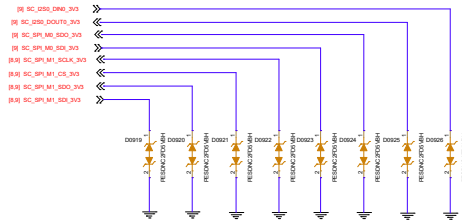


TVS



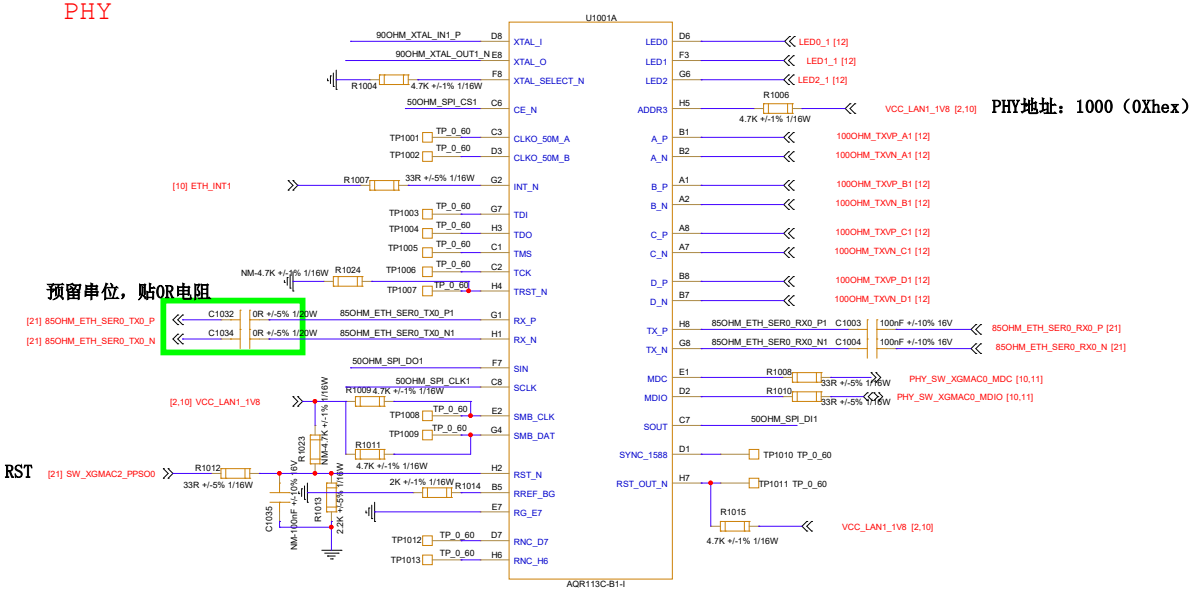






Quectel Wireless Solutions		
PROJECT: QSC801XP-AP-EVB		VER: V1.2
DRAWN BY: Cyrus	CHECKED BY: Terence	SIZE: A1
DATE: Monday, May 04, 2020	SHEET: 8 OF 21	

PHY

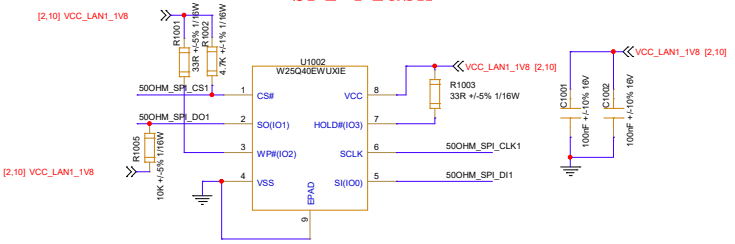


PHY地址：1000（0Xhex）

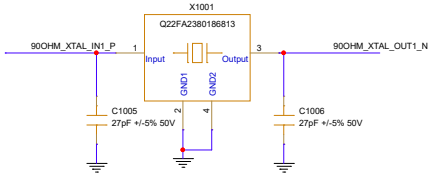
预留串位，贴0R电阻

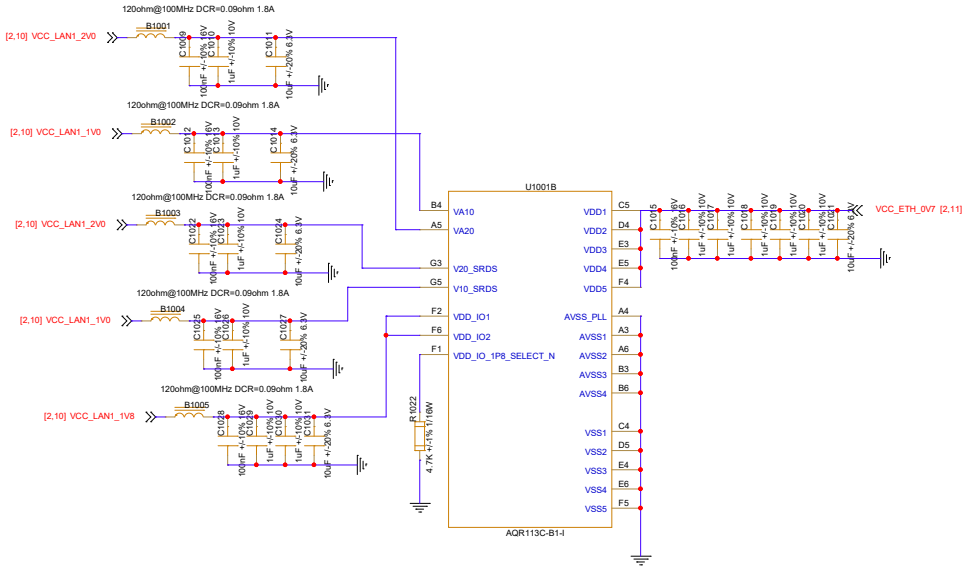
需要烧录固件，建议贴片前烧录

SPI Flash

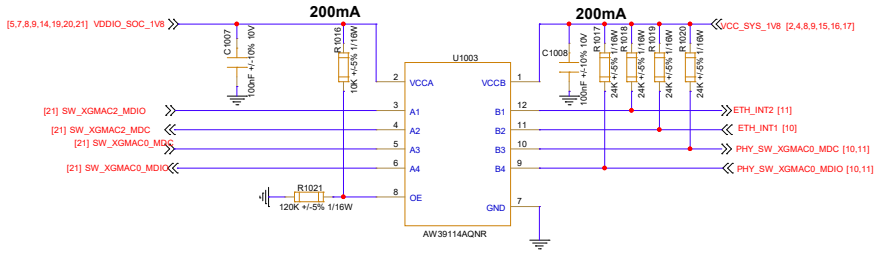


Crystal Oscillator

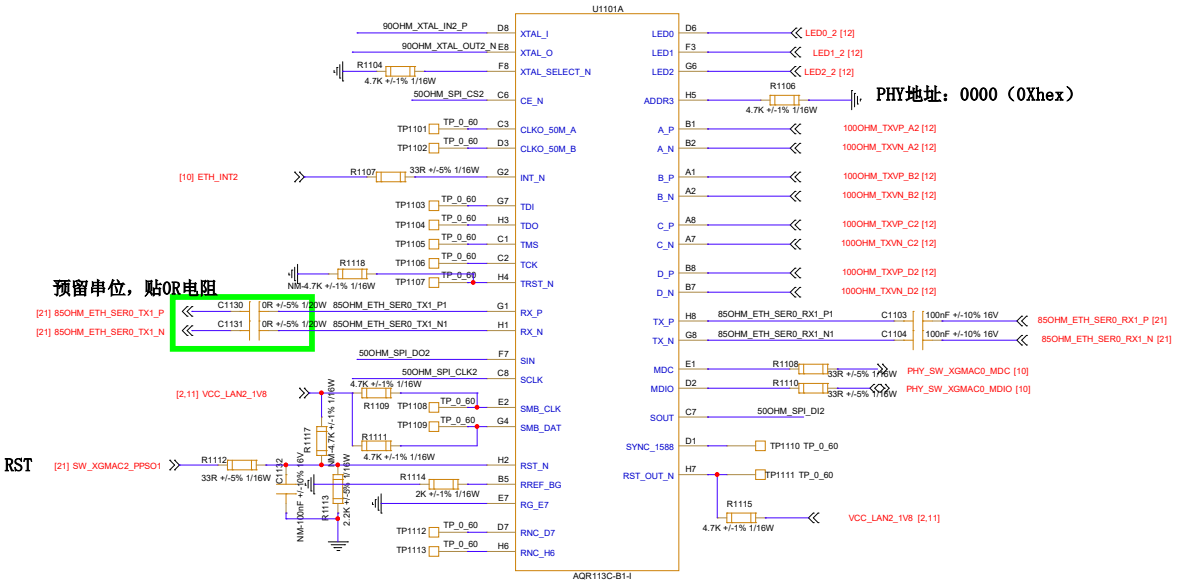




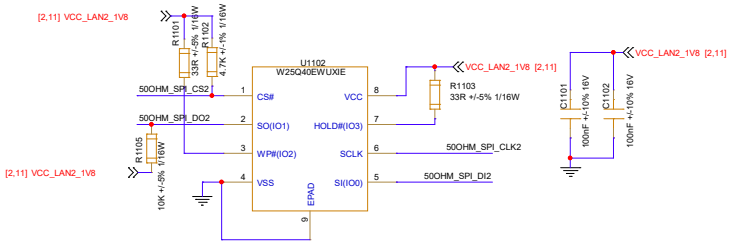
LEVEL SHIFT



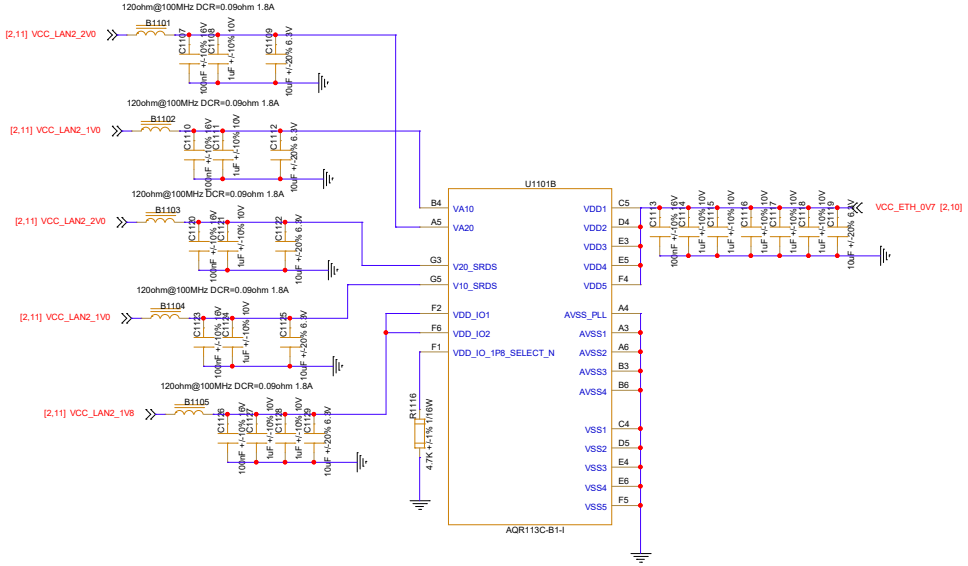
Quectel Wireless Solutions		
PROJECT: QSC801XP-AP-EVB		VER: V1.2
DRAWN BY: Cyrus	CHECKED BY: Terence	SIZE: A2
DATE: Monday, May 04, 2026	SHEET: 10OF 21	



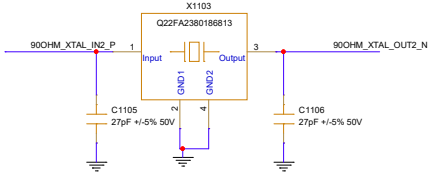
SPI Flash



需要烧录固件，建议贴片前烧录

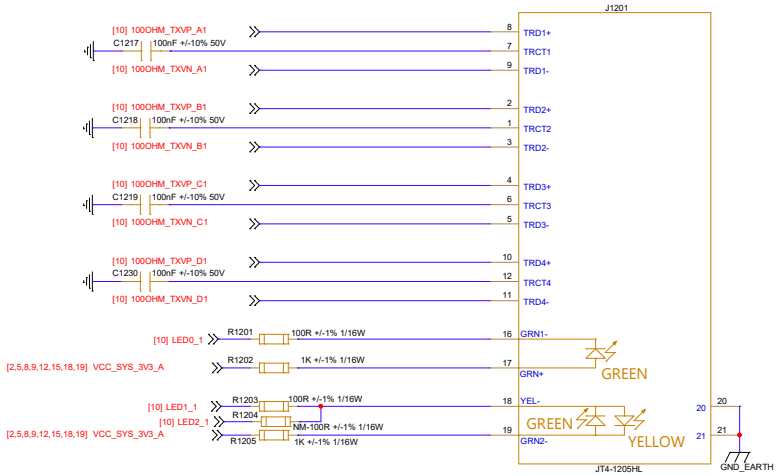


Crystal Oscillator

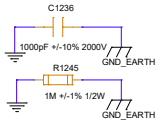


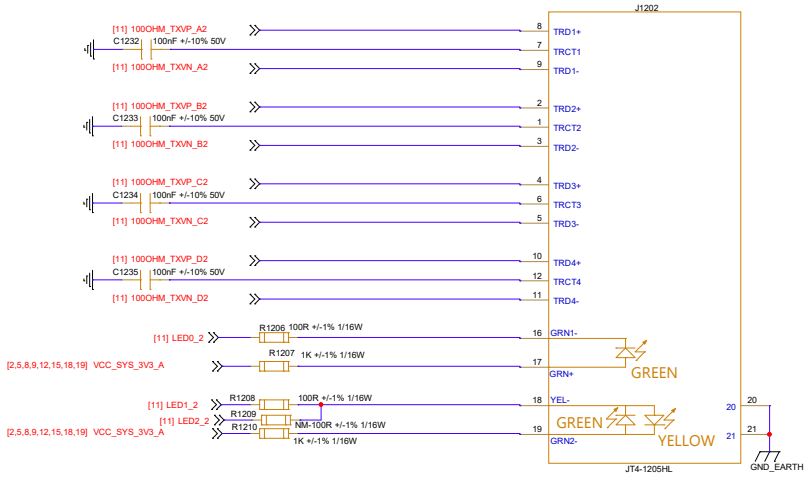
Quectel Wireless Solutions		
PROJECT: QSC801XP-AP-EVB		VER: V1.2
DRAWN BY: Cyrus	CHECKED BY: Terence	SIZE: A2
DATE: Monday, May 04, 2026	SHEET 11 OF 21	

Traditional Transformer 1



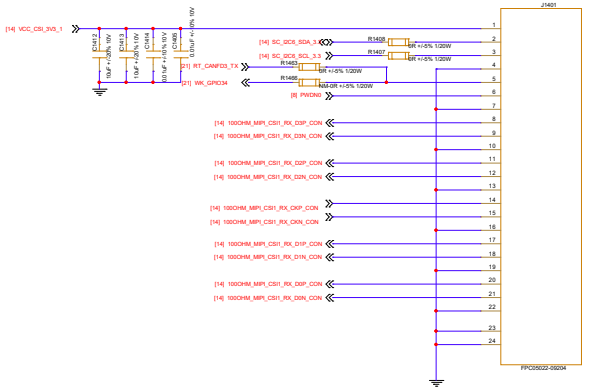
Traditional Transformer 2



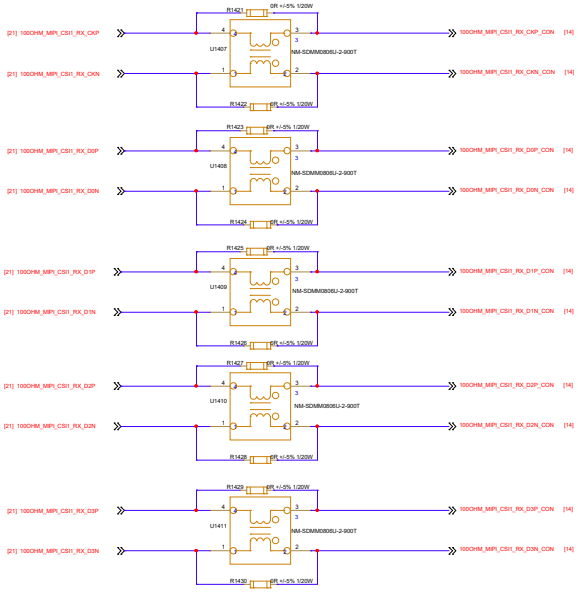


Quectel Wireless Solutions		
PROJECT: QSC801XP-AP-EVB		VER: V1.2
DRAWN BY: Cyrus	CHECKED BY: Terence	SIZE: A2
DATE: Monday, May 04, 2026	SHEET: 12OF 21	

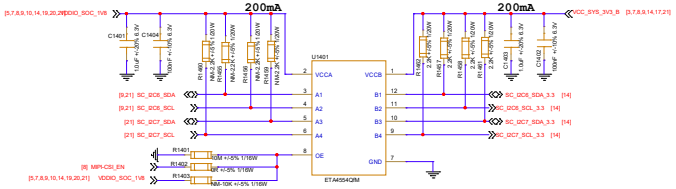
MIPI_CSI_1



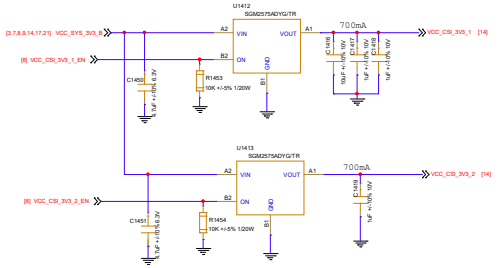
EMC Protection



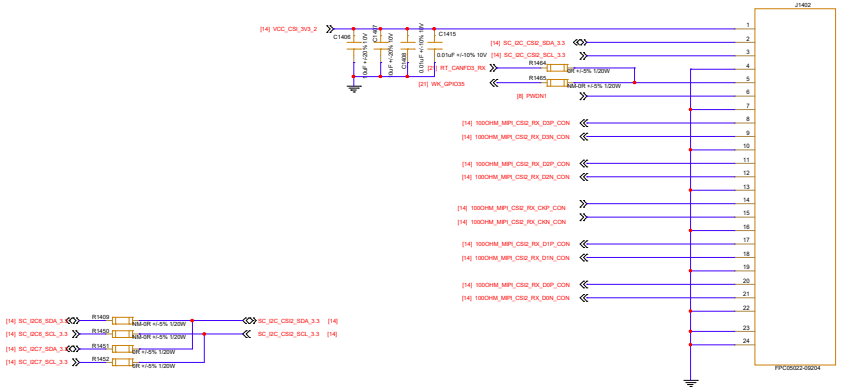
LEVEL SHIFT



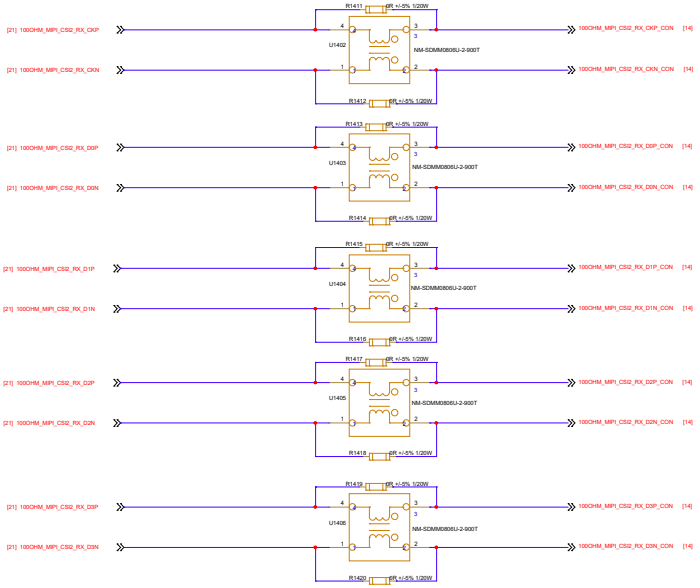
3.3V Power



MIPI_CSI_2

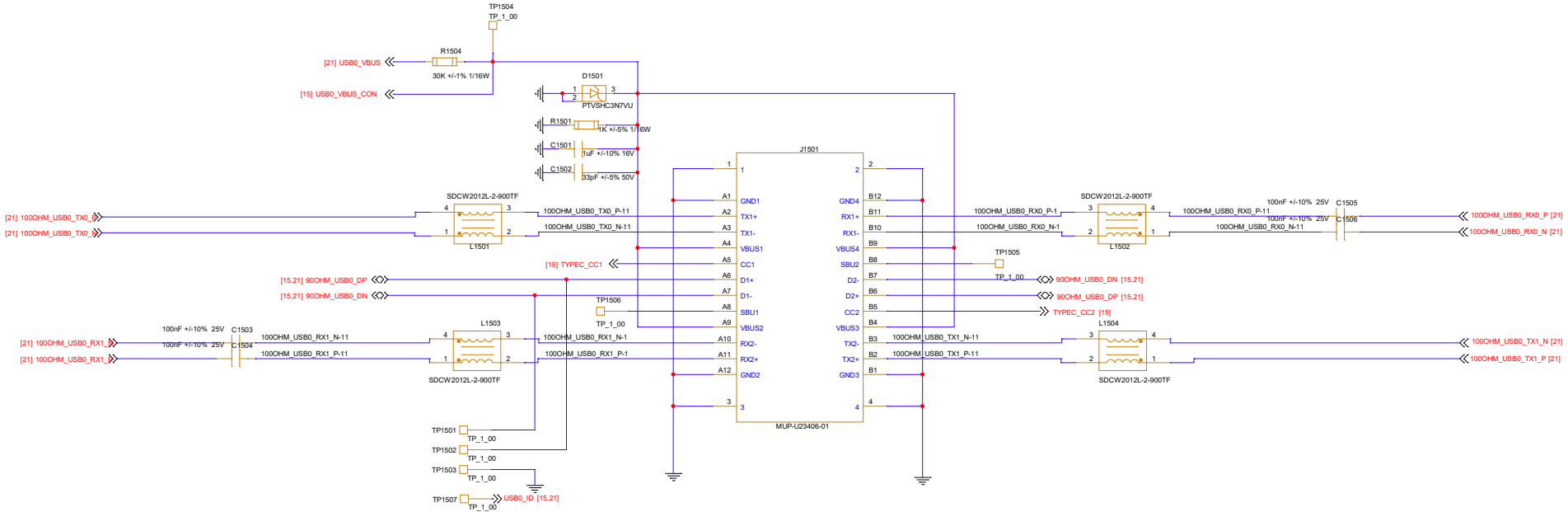


EMC Protection

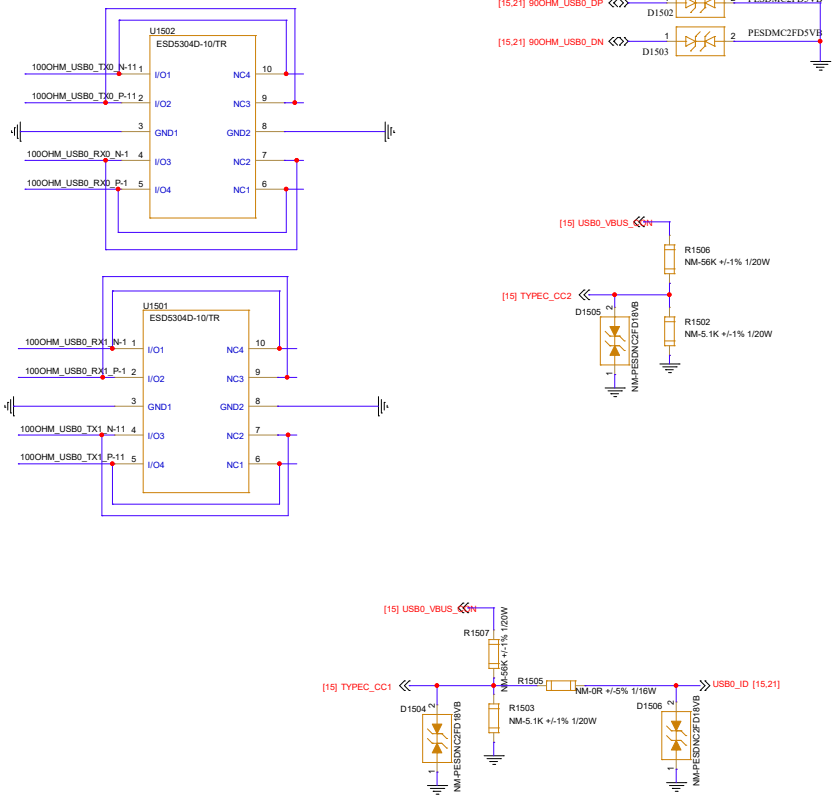


Quectel Wireless Solutions			
PROJECT	QSC8010XP-AP-EVB	VER	V1.2
DRAWN BY	Cyrus	CHECKED BY	Terence
SIZE	A1		
DATE	Monday, May 04, 2026	SHEET	1 OF 21

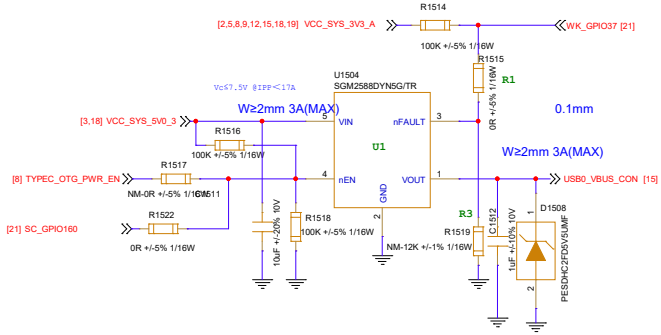
USB0-OTG



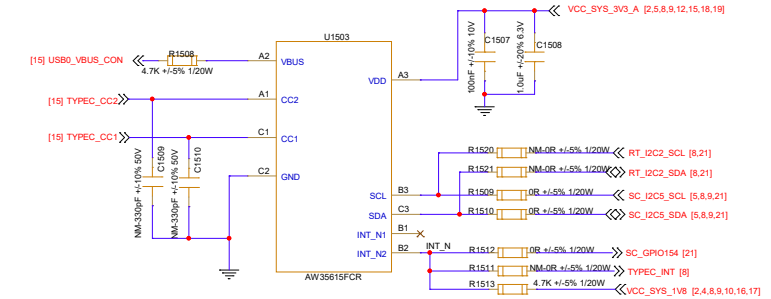
ESD



Over Current Protection Switch



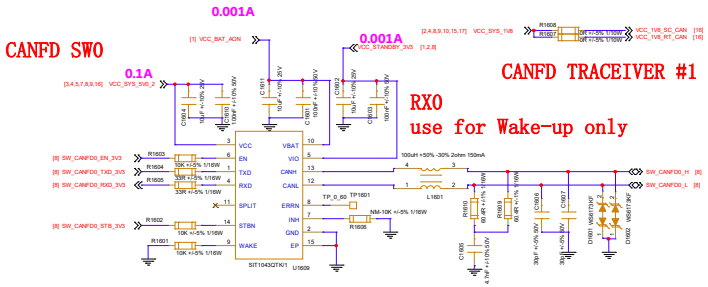
CC LOGIC



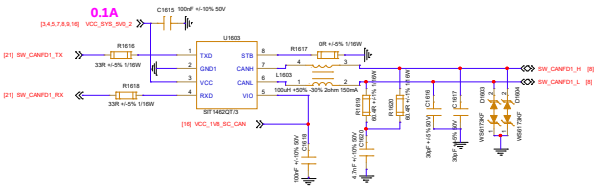
Quectel Wireless Solutions

PROJECT	QSC801XP-AP-EVB	VER	V1.2
DRAWN BY	Cyrus	CHECKED BY	Terence
SIZE	A2		
DATE	Monday, May 04, 2026	SHEET	15 OF 21

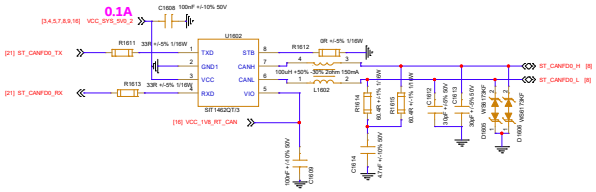
CANFD SW0



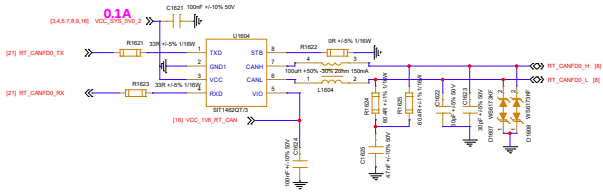
CANFD SW1



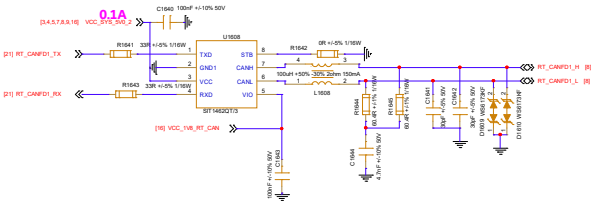
CANFD ST0



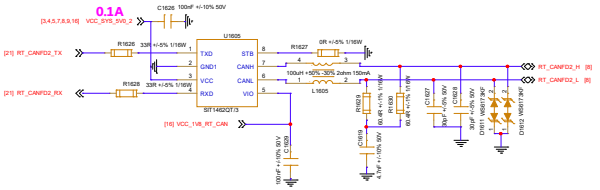
CANFD RT0



CANFD RT1

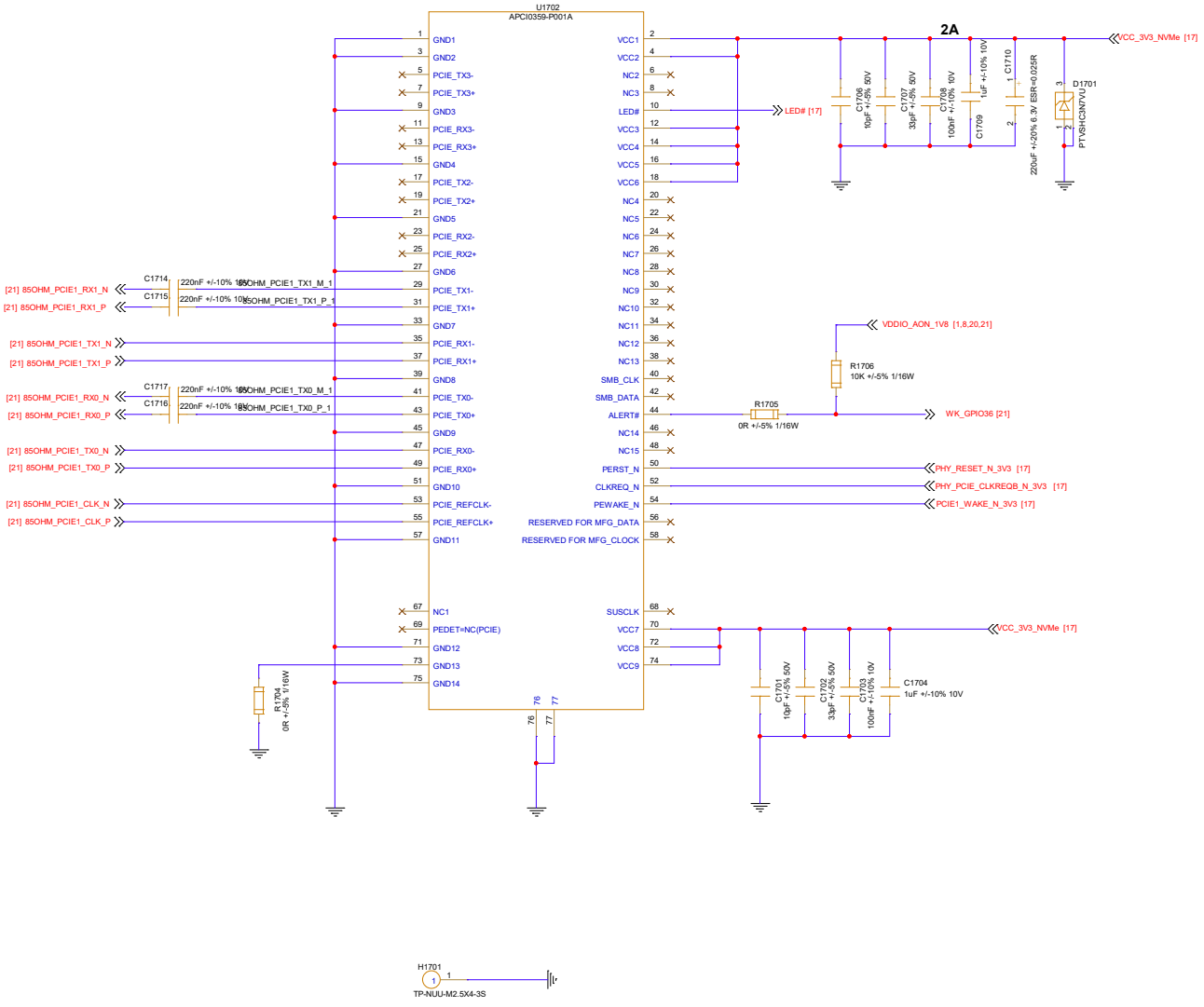


CANFD RT2

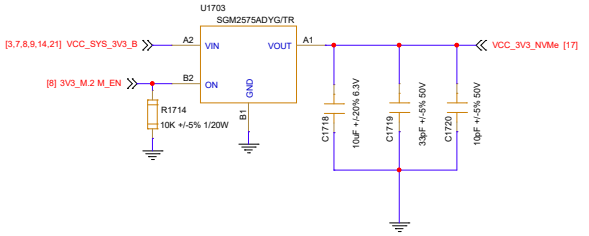


Quectel Wireless Solutions			
PROJECT	QSC801XP-AP-EVB	VER	V1.2
DRAWN BY	Cyrus	CHECKED BY	Terence
SIZE	A0		
DATE	Monday, May 04, 2026	SHEET	16 OF 21

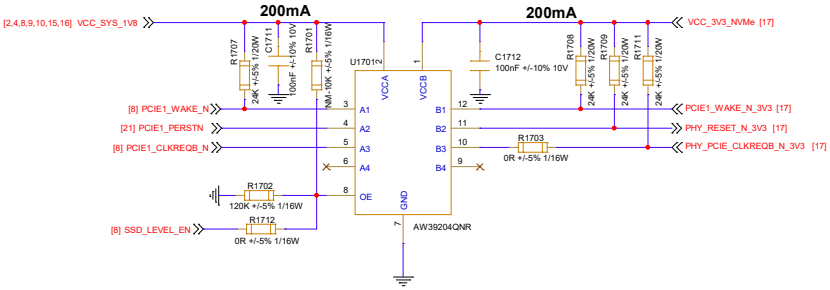
M.2 KEY M



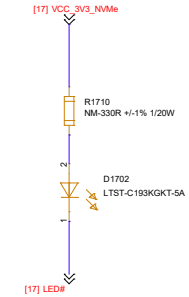
3.3V POWER



LEVEL SHIFT

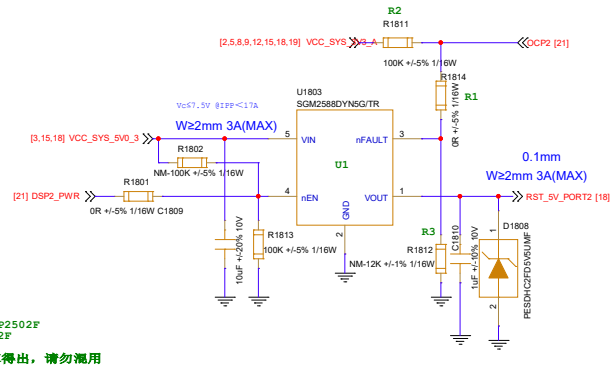


POWER LED



Quectel Wireless Solutions		
PROJECT: QSC801XP-AP-EVB		VER: V1.2
DRAWN BY: Cyrus	CHECKED BY: Terence	SIZE: A2
DATE: Monday, May 04, 2026	SHEET: 17OF 21	

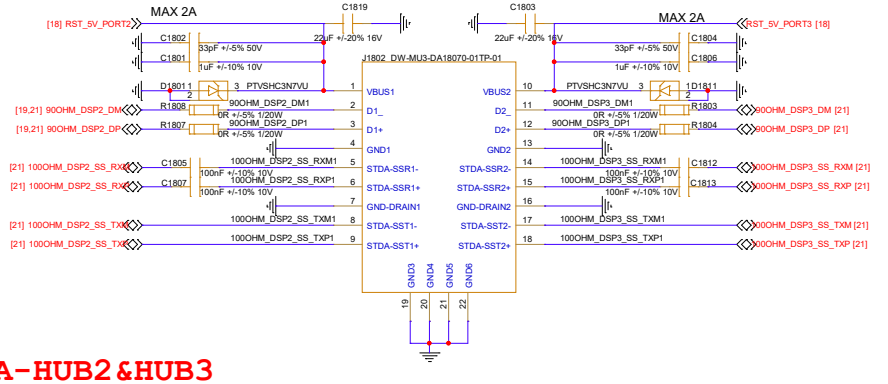
Over Current Protection Switch



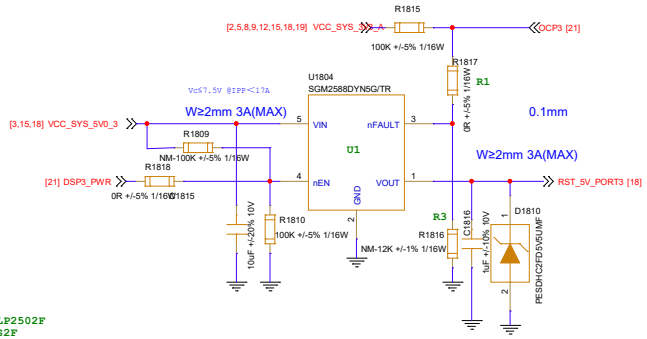
上件方案　　适用IC
方案1：U1+R1+R2　SGM2588,SGM2594,LP2502F
方案2：U1+R3　LP2502A,ETA6067GS2F

请注意，R3的阻值需要根据IC需求计算得出，请勿混用

TYPEA-HUB2&HUB3



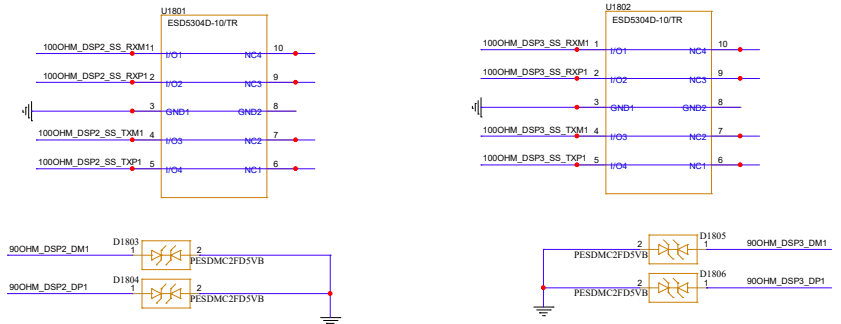
Over Current Protection Switch



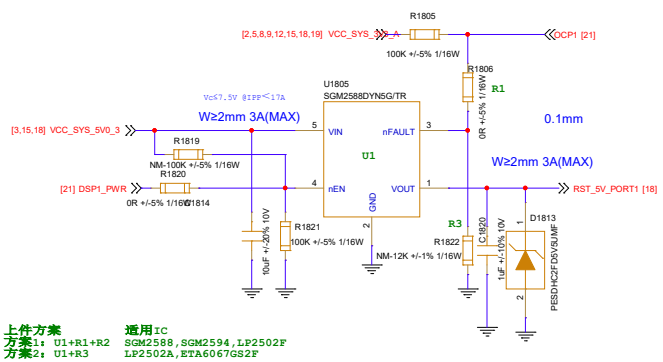
上件方案　　适用IC
方案1：U1+R1+R2　SGM2588,SGM2594,LP2502F
方案2：U1+R3　LP2502A,ETA6067GS2F

请注意，R3的阻值需要根据IC需求计算得出，请勿混用

ESD



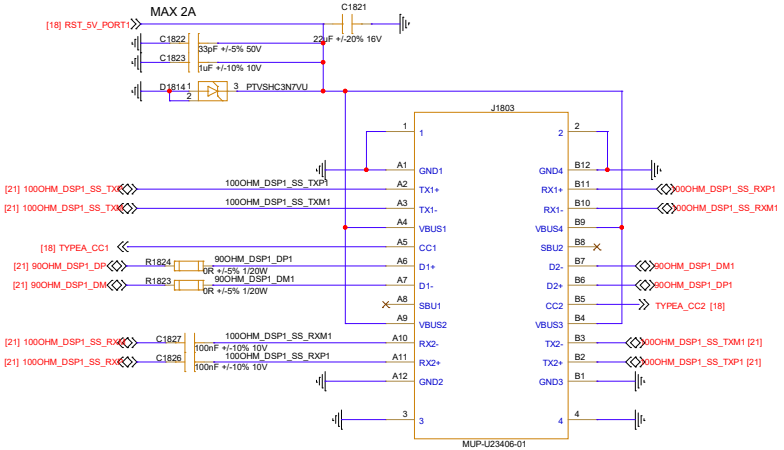
Over Current Protection Switch



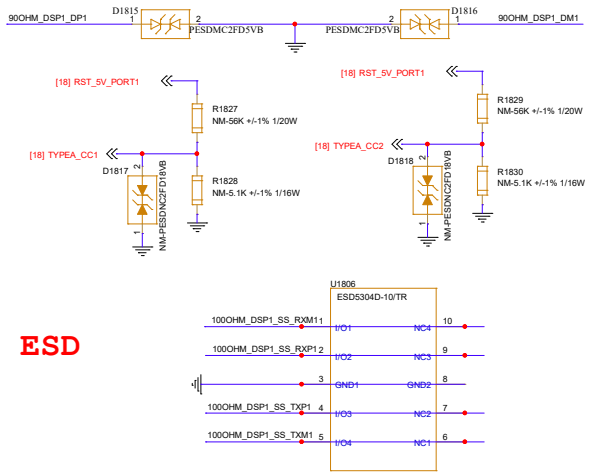
上件方案　　适用IC
方案1：U1+R1+R2　SGM2588,SGM2594,LP2502F
方案2：U1+R3　LP2502A,ETA6067GS2F

请注意，R3的阻值需要根据IC需求计算得出，请勿混用

TYPEC-HUB1

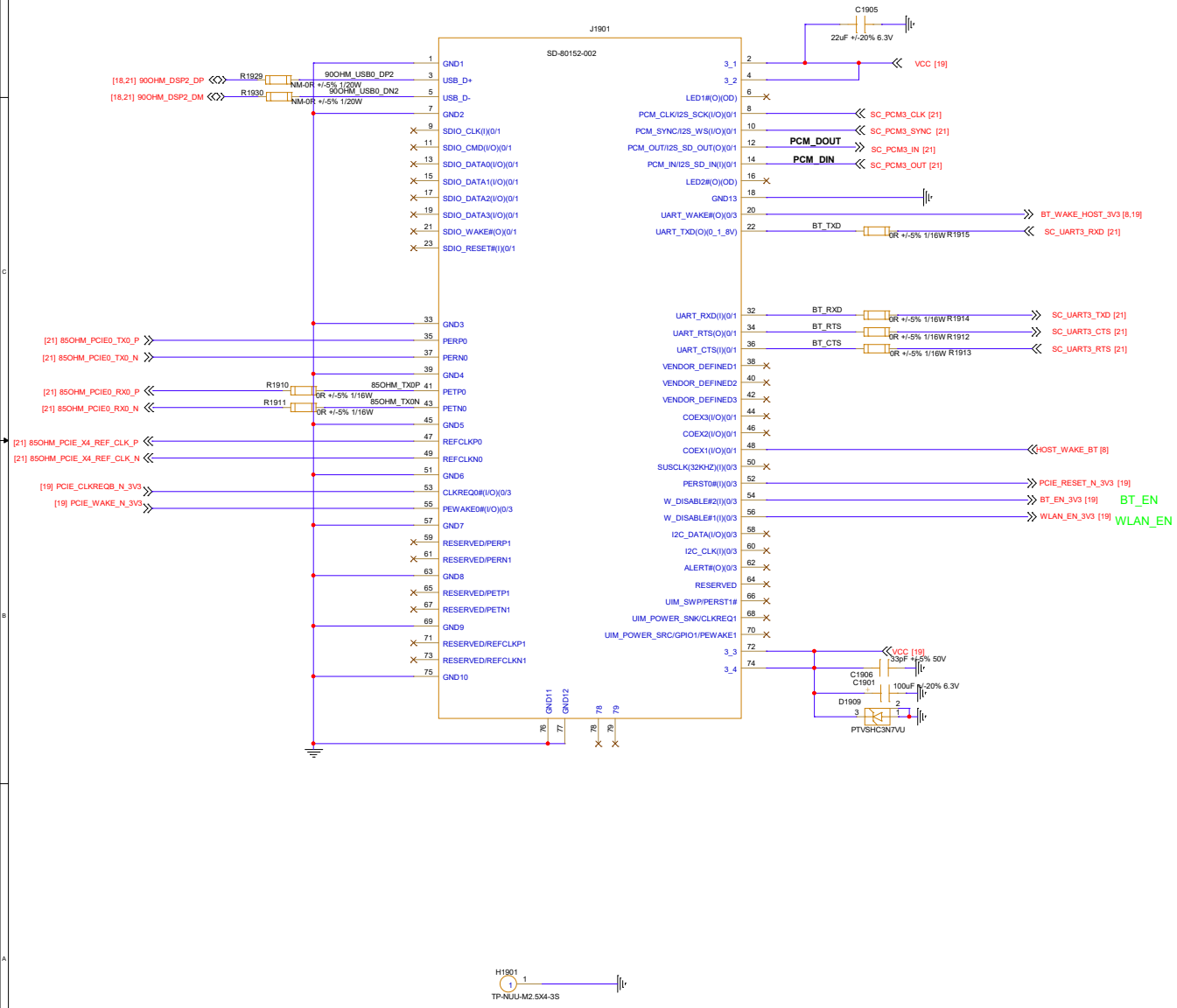


ESD



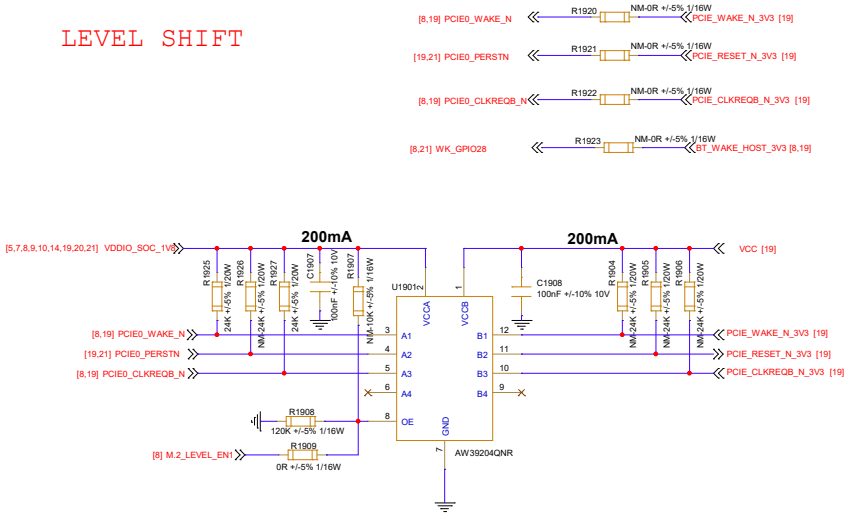
Quectel Wireless Solutions		
PROJECT	QSC801XP-AP-EVB	VER V1.2
DRAWN BY Cyrus	CHECKED BY Terence	SIZE A2
DATE Monday, May 04, 2026	SHEET	18OF 21

M.2 KEY E

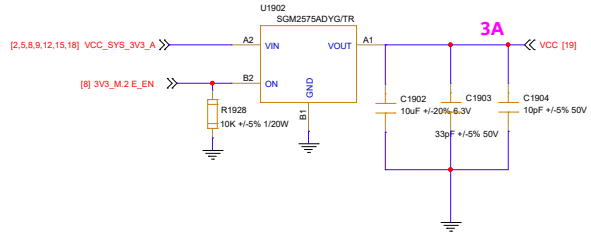


H1901
TP-NUU-M2.5X4-3S

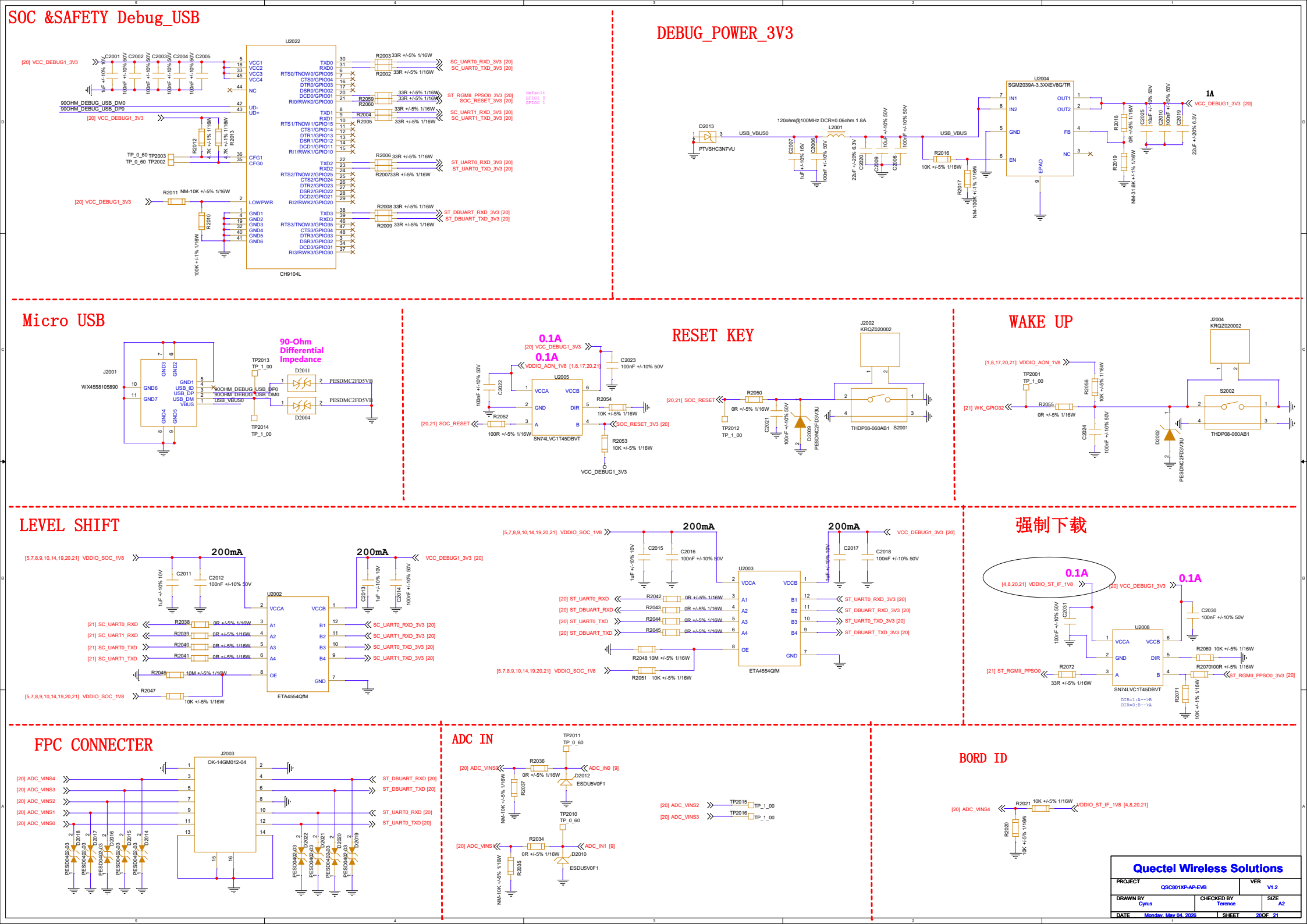
LEVEL SHIFT

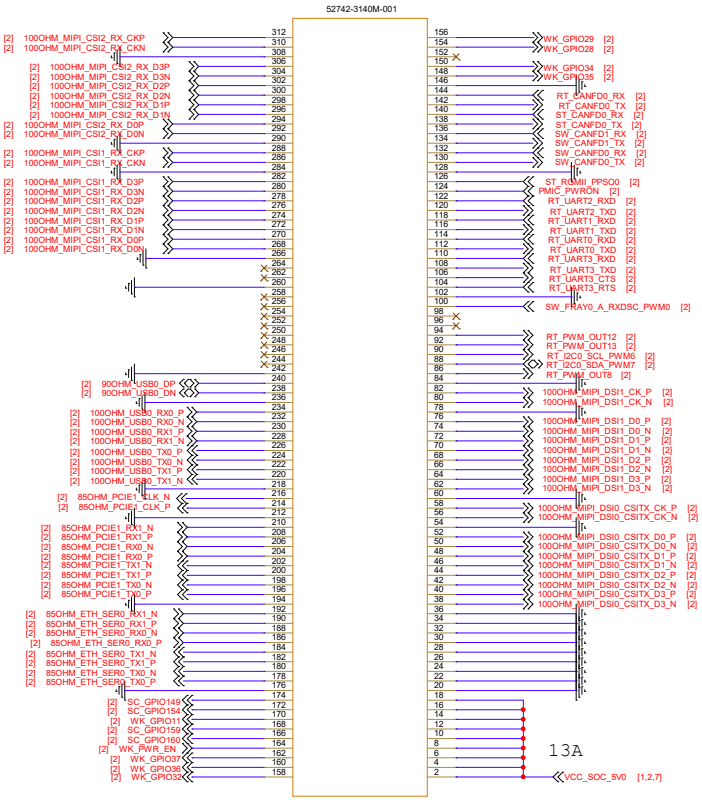


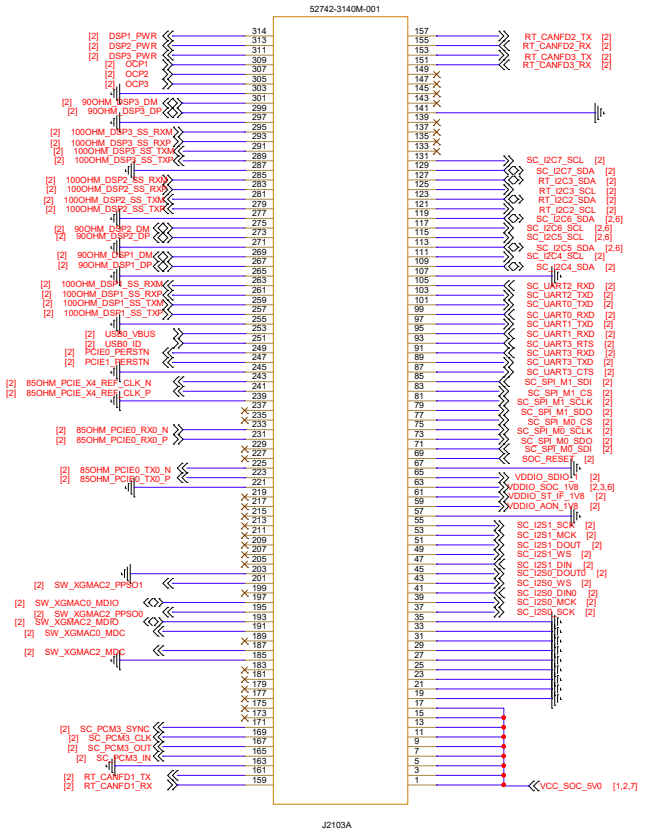
3.3V POWER

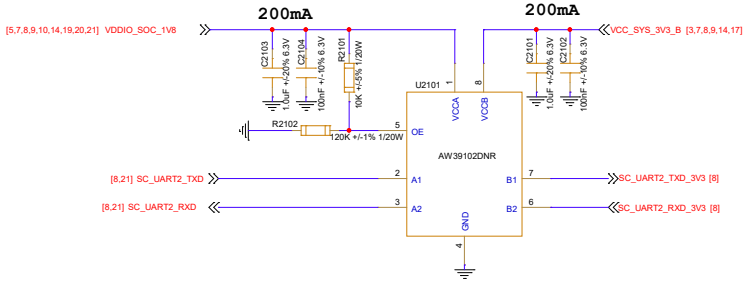


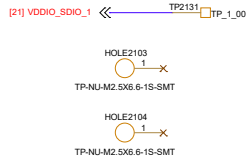
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DATE: Monday, May 04, 2026	SHEET: 19OF 21	


SOC &SAFETY Debug_USB
DEBUG_POWER_3V3
Micro USB
RESET KEY
WAKE UP
LEVEL SHIFT
强制下载
FPC CONNECTER
ADC IN
BORD ID
Quectel Wireless Solutions
PROJECT
QSC801XP-AP-EVB
VER
V1.2
DRAWN BY
Cyrus
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Terence
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